

Global, International, and Local CAPM

The standard capital asset pricing model (CAPM), introduced in Chapter 15, for estimating the cost of capital, does not explicitly account for foreign assets, foreign investors, or currencies. This raises the question whether such a model can provide the right cost of capital for investments in foreign currencies. If foreign-currency rates are changing, the same investment will generate different returns to investors from different countries. Take the case of a German government bond denominated in euros. From the perspective of a German or Dutch investor, this bond generates a risk-free return (assuming there is no inflation), because the euro is also the investor's domestic currency. But the bond's return is not risk free for investors in the United States, because the return measured in U.S. dollars will vary with the dollar-to-euro exchange rate.

As a general rule, investors from countries with different currencies are likely to disagree about an asset's expected return and risk. In theory, this means that the standard CAPM no longer holds, and a more complex, international CAPM is required. In practice, however, we find that the CAPM-based approach as laid out in Chapter 15 is still valid for estimating the cost of capital for cross-border investments. This appendix provides further background for our recommendations and practical guidelines for estimating the cost of capital in foreign currency.

GLOBAL CAPM

Investors' disagreement about the return and risk of international investments disappears if purchasing power parity (PPP) holds across all currencies. In that case, changes in exchange rates perfectly match differences in inflation between currencies:¹

$$X_t = X_{t-1} \left(\frac{1 + i_A}{1 + i_B} \right)$$

where X_t = exchange rate of currency B expressed in units of currency A at time t

i_A, i_B = inflation rate for currency A, B

As a result, the expected return and risk in real terms for any asset will be the same for all investors, regardless of their domestic currency. In the German bond example, any appreciation of the U.S. dollar relative to the euro would make the nominal bond return for U.S. investors lower. But if PPP holds, the inflation rate in the United States would be lower by exactly the same amount, so the payoff in real terms for U.S. and German investors would be equal. In real terms, there is no currency risk for investors. They will all hold the same global market portfolio of risky assets and face the same real risk-free rate as if there were only a single currency.

The resulting so-called global CAPM is in fact the standard CAPM with a global market portfolio. It expresses the expected real return for an asset j as follows:

$$E(r_j) = r_f + \beta_{j,G} [E(r_G) - r_f]$$

where r_j = return for asset j

r_f = risk-free rate

$\beta_{j,G}$ = beta of asset j versus global market portfolio G

r_G = return for global market portfolio G

According to the global CAPM, the cost of capital for domestic as well as foreign assets follows from the asset's beta relative to the global market portfolio and the market risk premium of that market portfolio relative to the risk-free rate.

Technically, the global CAPM is valid only if PPP holds. Evidence on PPP has been mixed, but academic research appears to conclude that deviations

¹ Technically, this is so-called relative purchasing power parity, referring to changes in prices and exchange rates. Absolute purchasing power parity requires that prices be the same across currencies; see, for example, R. Brealey, S. Myers, and F. Allen, *Principles of Corporate Finance*, 13th ed. (New York: McGraw-Hill, 2020), chap. 27.

from PPP between currencies are typically reduced to half their value within three to five years.² In other words, exchange rates do adjust for differences in inflation between countries, although not immediately and perfectly.

For investors and companies able to invest outside their home markets without restrictions, we recommend using the global CAPM to estimate the cost of capital for foreign as well as domestic investments. Effectively, this means applying the approach described in Chapter 15. Although the alternative, international CAPM (discussed next), may be theoretically superior, it is far more complex and does not lead to materially different results in practice.

INTERNATIONAL CAPM

If PPP does not hold, real returns from foreign assets are no longer free from currency risk, because changes in exchange rates are not offset by differences in inflation. The greater the correlation between the return on a foreign asset and the relevant currency rate, the higher the risk for an investor. Take, for example, a Dutch company whose stock returns, measured in euros, tend to be higher when the euro appreciates against the U.S. dollar and vice versa (for instance, because the company imports components from the United States and sells end products in Europe). The stock's returns will be riskier for an American investor than for a European investor, because the exchange rate tends to amplify the returns when translated into U.S. dollars. The absence of PPP means that disparities between dollar and euro inflation will not offset this difference in returns when measured in real terms.

To hold foreign assets, rational investors will require some compensation in the form of a higher expected return for an asset, depending on its exposure to currency risk. As a result, what matters for an asset's expected return is no longer only the asset's beta versus the global market portfolio (as in case of the global CAPM). The international CAPM captures the additional return requirements by also including asset betas versus currency exchange rates. For example, in a world consisting of three countries, each with its own currency, the international CAPM would define the expected return on asset j in a given home currency as follows:³

$$E(r_j) = r_f + \beta_{j,G} [E(r_G) - r_f] + \beta_{j,A} \text{CRP}_A + \beta_{j,B} \text{CRP}_B \quad (\text{G.1})$$

² For an overview, see A. M. Taylor and M. P. Taylor, "The Purchasing Power Parity Debate," *Journal of Economic Perspectives* 18, no. 4 (Fall 2004): 135–158.

³ This is a simplified version of the Solnik-Sercu international CAPM; see, for example, P. Sercu, *International Finance* (Princeton, NJ: Princeton University Press, 2009), chap. 19; and S. Armitage, *The Cost of Capital* (Cambridge: Cambridge University Press, 2005), chap. 11.

where r_j = return for asset j
 r_f = risk-free rate
 $\beta_{j,G}$ = beta of asset j versus global market portfolio G
 $\beta_{j,A}, \beta_{j,B}$ = beta of asset j versus currency rate X_A, X_B
 CRP_A, CRP_B = risk premium for currency A, B

The currency risk premiums are defined as follows:

$$CRP_n = \frac{E(X_{n1}) - F_{n1}}{X_{n0}} \quad (G.2)$$

where X_{nt} = exchange rate of home currency expressed in units of currency n at time t where $n = A, B$
 F_{n1} = forward rate for $t = 1$ of home currency expressed in units of currency n

Although theoretically correct, the international CAPM is probably too cumbersome for practical use. In particular, it is not clear how many of the world's currencies to include in estimating the cost of capital. Even taking only a handful of leading global currencies would require that you estimate as many currency risk premiums. Further, in addition to an asset's market beta, you would need to estimate its beta versus each of these currencies.

Another reason not to use the international CAPM is that empirical research has shown that the currency risk premiums are typically too small to matter when estimating a cost of capital.⁴ According to recent research that compared cost of capital estimates from a global and an international CAPM for large U.S. companies, differences are probably less than half a percentage point.⁵ As we can see from Equations G.1 and G.2, the international CAPM simplifies to the global CAPM when currency risk premiums are negligible. In other words, PPP apparently holds sufficiently well for the global CAPM to lead to the same cost of capital as the international CAPM. Expressed either way, this evidence reinforces our recommendation to use the global CAPM.

⁴ Sercu, *International Finance*, chap. 19.

⁵ See W. Dolde, C. Giaccotto, D. Mishra, and T. O'Brien, "Should Managers Estimate Cost of Equity Using a Two-Factor International CAPM?" *Managerial Finance* 38, no. 8 (2012): 708–728; and D. Mishra and T. O'Brien, "A Comparison of Cost of Equity Estimates of Local and Global CAPMs," *Financial Review* 36, no. 4 (2001): 27–48.

LOCAL CAPM

Some practitioners and academic researchers propose estimating the cost of capital for an investment opportunity in a particular country by using a local CAPM. The investment's beta is then estimated versus the market portfolio of the country, and the market risk premium follows from the excess return of that same market portfolio over the local risk-free rate. The approach is theoretically correct if stocks are correlated to the global market portfolio only through the local market:⁶

$$\beta_{j,G} = \beta_{j,L} \times \beta_{L,G} \quad (\text{G.3})$$

where $\beta_{j,G}$ = beta of asset j versus global market portfolio G

$\beta_{j,L}$ = beta of asset j versus local market portfolio L

$\beta_{L,G}$ = beta of local market portfolio L versus global market portfolio G

This implies that any international risk factors influencing the returns of companies in a given country are fully captured by the local market portfolio of that country. You can then indirectly estimate any asset's global beta by multiplying its local beta by the global beta of the local market. If the local stock market is fully integrated and correctly priced in the global market, its expected return is:

$$E(r_L) = r_f + \beta_{L,G}[E(r_G) - r_f] \quad (\text{G.4})$$

where r_L = expected return for local market portfolio L

r_f = risk-free rate

r_G = return for global market portfolio G

Combining Equations G.3 and G.4 shows that the expected return for a stock j estimated via the local and global CAPM should be equal as well. Following the global CAPM, this return is given by:

$$E(r_j) = r_f + \beta_{j,G}[E(r_G) - r_f]$$

Substituting the asset's global beta by the indirect beta defined previously in Equation G.3 leads to:

$$E(r_j) = r_f + \beta_{j,L} \times \beta_{L,G}[E(r_G) - r_f]$$

⁶ See R. Stulz, "The Cost of Capital in Internationally Integrated Markets: The Case of Nestlé," *European Financial Management* 1, no. 1 (1995): 11–22.

This can be rearranged to show equivalence with the local CAPM:

$$E(r_j) = r_f + \beta_{j,L}[E(r_L) - r_f]$$

Although the assumptions may not seem very realistic at face value, there is evidence that the local and global CAPM generate similar results. Empirical research finds that the cost of capital estimated for U.S. companies with a local CAPM is very close to the estimate based on a global CAPM.⁷ For U.S. stocks, this may not be surprising, as the U.S. market portfolio is well diversified and highly correlated with the global market portfolio. But supporting evidence also comes from nine developed economies, including not only the United States but also the United Kingdom, Germany, France, and smaller economies such as the Netherlands and Switzerland. An analysis of beta estimates for companies versus a local and global market portfolio has shown that for these countries, the betas are typically related, as indicated by Equation G.3.⁸

However, the local CAPM approach, when compared with the global CAPM, has some practical drawbacks. First is that when you apply the local CAPM to investments in different countries, you should estimate the local market risk premium and beta for each of these countries, instead of only the global market risk premium, as you would do when applying the global CAPM. Also, with a local CAPM, you cannot make a straightforward estimate of a company's beta based on the average of the estimated betas for a sample of industry peers (which Chapter 15 recommends to reduce the standard error of the company's beta). The reason is that if the peers are in different countries, their local betas are not directly comparable. Finally, local risk premiums are typically less stable over time than their aggregate, the global risk premium. For example, Exhibit G.1 compares the realized premiums on local stock market indexes with government bond returns for several countries and a globally diversified portfolio, using data from Dimson, Marsh, and Staunton's analysis of long-term average returns on equities and corporate and government bonds.⁹ The individual countries' risk premiums vary considerably, depending on the time period over which they are measured, while the global premium remains almost unchanged.

Note that the risk premium differences shown in Exhibit G.1 do not mean that the price for risk varies across these countries. These differences are driven by several factors. First, levels of economic development and, therefore, profit

⁷ R. Harris, F. Marston, D. Mishra, and T. O'Brien, "Ex-Ante Cost of Equity Estimates of S&P 500 Firms: The Choice between Domestic and Global CAPM," *Financial Management* 32, no. 3 (2003): 51–66.

⁸ See C. Koedijk, C. Kool, P. Schotman, and M. van Dijk, "The Cost of Capital in International Financial Markets: Local or Global?," *Journal of International Money and Finance* 21, no. 6 (2002): 905–929.

⁹ E. Dimson, P. Marsh, and M. Staunton, *Triumph of the Optimists: 101 Years of Global Investment Returns* (Princeton, NJ: Princeton University Press, 2002); and E. Dimson, P. Marsh, M. Staunton, and J. Wilmot, *Credit Suisse Global Investment Returns Yearbook 2016* (London: Credit Suisse Research Institute, 2016).

EXHIBIT G.1 Comparing Risk Premiums across Countries and over Time

Annualized market risk premium over 1-year Treasury bills, %



Source: E. Dimson, P. Marsh, and M. Staunton, *Triumph of the Optimists: 101 Years of Global Investment Returns* (Princeton, NJ: Princeton University Press, 2002); E. Dimson, P. Marsh, M. Staunton, and J. Wilmot, *Credit Suisse Global Investment Yearbook 2016* (London: Credit Suisse Research Institute, February 2016).

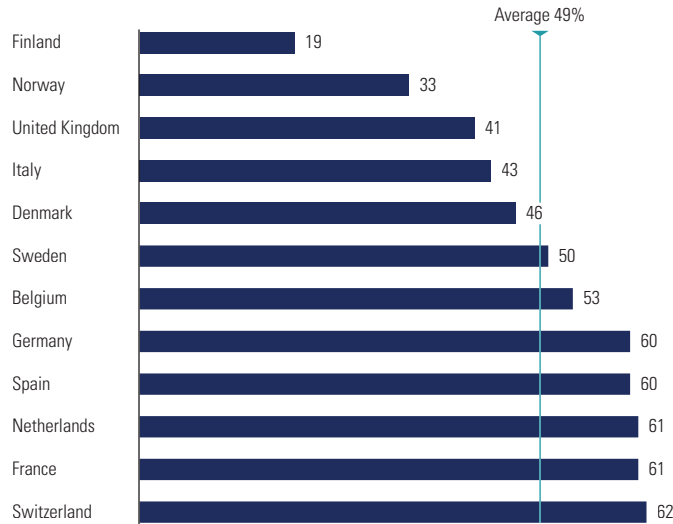
¹ Globally diversified portfolio.

growth have varied over the past century among the countries. Second, capital markets were less integrated in the past, so prices across countries may not have been equalized. The main reason, though, is that many of the stock market indexes used had different levels of diversification and beta. Therefore, their performance was skewed by different industry concentrations. In most European countries, the key stock market indexes, which account for the majority of their stock markets' total capitalization, typically include only 25 to 40 companies, often from a limited range of industries. Indeed, research has shown that a large fraction of the variation in returns on European market indexes could be explained by their industry composition (see Exhibit G.2).¹⁰

¹⁰ R. Roll, "Industrial Structure and the Comparative Behavior of International Stock Market Indexes," *Journal of Finance* 47, no. 1 (1992): 3–42.

EXHIBIT G.2 **Share of Equity Returns Explained by Industry Composition of Index**

Adjusted R^2 , %



Source: R. Roll, "Industrial Structure and the Comparative Behaviour of International Stock Market Indexes," *Journal of Finance* 47, no. 1 (1992): 3–42.

We recommend a local CAPM only for investors and companies facing restrictions on investing abroad. In that case, the local market portfolio is the right reference to estimate the cost of capital. As a result, valuations in such restricted markets can be out of line with those in global markets—which is what we have encountered in the past for valuations in, for example, Asian stock markets.